

# Outcomes After Minor Ischemic Stroke in Older Patients Treated With IV Thrombolysis vs Standard of Care in the TEMPO-2 Trial

Aravind Ganesh,<sup>1,2</sup> Shabnam Vatanpour,<sup>1</sup> Amy Ying Xin Yu,<sup>3</sup> Philip Barber,<sup>1,4</sup> Philip Man Chiu Choi,<sup>5</sup> Thalia S. Field,<sup>6</sup> Mahesh P. Kate,<sup>7</sup> Timothy John Kleinig,<sup>8</sup> Michel Christopher Frank Shamy,<sup>9</sup> Kausik Chatterjee,<sup>10</sup> Carlos A. Molina,<sup>11</sup> Bruce C.V. Campbell,<sup>12</sup> Ramana Appireddy,<sup>13</sup> Keith W. Muir,<sup>14</sup> Michael D. Hill,<sup>1,2</sup> and Shelagh B. Coutts,<sup>1,2</sup> as the TEMPO-2 investigators

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## Correspondence

Dr. Ganesh  
aganesh@ucalgary.ca

## Abstract

### Background and Objectives

There are few high-quality data on thrombolysis outcomes in the oldest patients, especially for minor stroke. We examined safety and efficacy outcomes of thrombolysis in patients older than 80 years, compared with those  $\leq 80$  years, in the Tenecteplase Versus Standard of care for Minor Ischaemic Stroke With Proven Occlusion (TEMPO-2) trial.

### Methods

In this post hoc analysis of the TEMPO-2 randomized controlled trial, we compared outcomes and adverse events in patients with minor stroke and symptomatic intracranial occlusion or focal perfusion lesion within 12 hours of symptom onset, assigned to tenecteplase vs non-thrombolytic standard of care, in those  $>80$  years and  $\leq 80$  years. The primary outcome was responder analysis of the 90-day modified Rankin Scale (responder was mRS = 0–1 if pre-morbid mRS = 0–1, 0–2 if pre-morbid mRS = 2). Secondary outcomes included neurologic recovery according to the NIH Stroke Scale (NIHSS) at 5 days/discharge, vessel recanalization, Lawton Instrumental Activities of Daily Living Scale, EuroQol-5 Dimension (EQ-5D), and adverse events. We used mixed-effects Poisson, ordinal logistic, and quantile regression as appropriate, adjusted for sex, baseline NIHSS, and treatment assignment as fixed-effects and enrolling site as random effects.

### Results

Among 884 patients in the intention-to-treat analysis (208 [23.5%]  $>80$  years, 365 [41.6%] female). Significant interactions between age and treatment were observed for mRS responder, mRS 0–1, return to baseline function, EQ-5D index, and vessel recanalization. Patients  $>80$  years generally fared worse with tenecteplase for mRS outcomes (mRS 0–1 in 49 [46.2%] vs 61 [59.8%] with control, adjusted risk ratio [aRR] 0.83, 95% CI 0.72–0.97), while among patients  $\leq 80$  years, there was no significant difference (249 [76.4%] vs 260 [74.7%], aRR 1.01, 95% CI 0.98–1.03). However, in both age groups, patients were more likely to achieve recanalization of visible occlusions (aRR  $\leq 80$  years 2.06, 95% CI 1.60–2.65,  $>80$  years 2.77, 95% CI 2.21–3.47) and NIHSS = 0 at 5 days/discharge when assigned to tenecteplase (aRR  $\leq 80$  years 1.14, 95% CI 1.04–1.26,  $>80$  years 1.21, 95% CI 1.10–1.33). Serious adverse events (SAEs) were more frequent with tenecteplase among patients  $>80$  years (risk ratio 2.29, 95% CI 1.27–4.13), particularly hemorrhages and stroke progression/recurrence (any extra/intracranial hemorrhage in 5 [4.7%] vs 0 with standard of care,  $p = 0.026$ ).

## MORE ONLINE

### Class of Evidence

Criteria for rating therapeutic and diagnostic studies

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### Supplementary Material

<sup>1</sup>Calgary Stroke Program, Department of Clinical Neurosciences and the Hotchkiss Brain Institute, University of Calgary Cumming School of Medicine, Alberta, Canada; <sup>2</sup>Departments of Community Health Sciences, Medicine, and Radiology, and the O'Brien Institute for Public Health, University of Calgary Cumming School of Medicine, Alberta, Canada; <sup>3</sup>Department of Medicine (Neurology), University of Toronto, Sunnybrook Health Sciences Centre, Ontario, Canada; <sup>4</sup>Department of Radiology, University of Calgary Cumming School of Medicine, Alberta, Canada; <sup>5</sup>Department of Neuroscience, Box Hill Hospital, Eastern Health, Melbourne, Australia and Eastern Health Clinical School, Monash University, Melbourne, Australia; <sup>6</sup>Vancouver Stroke Program, Division of Neurology, University of British Columbia, Vancouver, Canada; <sup>7</sup>Division of Neurology, Department of Medicine, University of Alberta, Edmonton, Canada; <sup>8</sup>Department of Neurology, Royal Adelaide Hospital, and Department of Medicine, University of Adelaide, Australia; <sup>9</sup>Department of Medicine, University of Ottawa and Ottawa Hospital Research Institute, Ontario, Canada; <sup>10</sup>Countess of Chester Hospital NHS Foundation Trust, United Kingdom; <sup>11</sup>Vall d'Hebron Stroke Center, Hospital Vall d'Hebron, Barcelona, Spain; <sup>12</sup>Department of Medicine and Neurology, Melbourne Brain Centre at the Royal Melbourne Hospital, University of Melbourne, Parkville, Australia; <sup>13</sup>Division of Neurology, Department of Medicine, Queen's University, Kingston, Canada; and <sup>14</sup>School of Neuroscience & Psychology, University of Glasgow, United Kingdom.

## Glossary

**acOR** = adjusted common odds ratio; **ARAMIS** = Antiplatelet vs R-tPA for Acute Mild Ischemic Stroke; **aRR** = adjusted RR; **EQ-5D** = EuroQol-5 Dimension; **HR** = hazard ratio; **IADL** = Instrumental Activities of Daily Living; **IQR** = interquartile range; **mRS** = modified Rankin Scale; **NIHSS** = NIH Stroke Scale; **PRISMS** = Potential of rtPA for Ischemic Strokes With Mild Symptoms; **PUMICE** = Prourokinase for Mild Ischemic Cerebrovascular Events; **RR** = risk ratio; **SAE** = serious adverse event; **TEMPO-2** = Tenecteplase Versus Standard of care for Minor Ischaemic Stroke With Proven Occlusion; **TRUST** = Thrombolysis of Urokinase for minor STroke Trial.

## Discussion

Older patients (>80 years) with minor stroke and visible occlusion or perfusion lesion assigned to tenecteplase were more likely to achieve recanalization of occlusions and short-term neurologic recovery, as were younger patients, but they experienced worse 90-day outcomes with more frequent hemorrhagic and stroke progression/recurrence-related SAEs. Overall, these results, while post hoc, do not support thrombolysis for minor strokes in older patients.

## Trial Registration Information

ClinicalTrials.gov: NCT02398656.

## Classification of Evidence

This study provides Class III evidence that in patients >80 years with minor stroke and visible occlusion or perfusion lesion, IV thrombolysis with tenecteplase is associated with worse functional outcomes at 90 days compared with nonthrombolytic standard of care therapies.

## Introduction

Age is a critical determinant of stroke outcomes, with older individuals being at greater risk of experiencing severe disability and mortality.<sup>1-3</sup> Patients older than 80 years have often been excluded from randomized-controlled trials of thrombolysis<sup>4,5</sup>; nevertheless, IV thrombolysis is recommended in international guidelines for these patients when they present with acute ischemic stroke.<sup>6</sup>

However, data on thrombolysis in older patients are especially sparse for minor ischemic stroke, and even more so for thrombolysis with tenecteplase, which has now been established as a preferred alternative to alteplase.<sup>7</sup> The Potential of rtPA for Ischemic Strokes With Mild Symptoms (PRISMS) and Antiplatelet vs R-tPA for Acute Mild Ischemic Stroke (ARAMIS) trials both examined thrombolysis with IV alteplase in patients with minor stroke. Subgroup analyses by age in these trials compared the primary outcome of modified Rankin Scale (mRS) 0–1 at 90 days in patients aged ≥65 years (123/313 [39.3%] patients in PRISMS and 353/719 [49.1%] in ARAMIS) to those <65 years.<sup>8,9</sup> PRISMS was terminated very early and did not show benefit of alteplase over aspirin in either age group and in ARAMIS, dual antiplatelet therapy was noninferior to IV alteplase in both age groups. Recently, the Prourokinase for Mild Ischemic Cerebrovascular Events (PUMICE) trial also found that prourokinase was not superior to standard care in patients with minor stroke, including in patients ≥80 years who constituted a very small proportion of the population (93/1,445 [6.4%]).<sup>10</sup> Similarly, the Thrombolysis of Urokinase for minor STroke Trial (TRUST)

found that IV urokinase was not beneficial for improving rates of excellent functional outcome at 90 days in patients with acute minor stroke treated within 6 hours of symptom onset, including in patients ≥65 years (496/999 [49.6%]); this trial excluded patients who were aged >80 years.<sup>11</sup>

The Tenecteplase Versus Standard of care for Minor Ischaemic Stroke With Proven Occlusion (TEMPO-2) randomized phase 3 clinical trial compared tenecteplase vs standard of care treatment for patients with minor ischemic stroke with a symptomatic intracranial occlusion or focal perfusion lesion.<sup>12</sup> The trial, which did not show superiority of tenecteplase to standard of care, enrolled many older patients (23% of participants). Therefore, we undertook a detailed examination of the safety and efficacy outcomes of thrombolysis vs standard of care in patients >80 years, compared with those ≤80 years. The primary research question examined was whether IV thrombolysis with tenecteplase is associated with better functional outcomes at 90 days than nonthrombolytic standard of care therapies in patients >80 years of age with minor stroke and visible occlusion or perfusion lesion.

## Methods

TEMPO-2 was a randomized, multicenter, phase 3 clinical trial, open label with blinded outcome assessment, that tested the superiority of IV tenecteplase (0.25 mg/kg) over nonthrombolytic standard of care in patients with minor ischemic stroke deficits and symptomatic intracranial occlusion or focal perfusion lesion within 12 hours of symptom onset. The trial protocol<sup>13</sup> and main results<sup>12</sup> have been published. In brief,

eligibility criteria included adults  $\geq 18$  years (no upper age limit), baseline independence (mRS  $\leq 2$ ), minor deficits (NIH Stroke Scale [NIHSS] score  $\leq 5$ ), and no evidence of evolved infarction concordant with the acute presenting syndrome or Alberta Stroke Program Early CT score  $\geq 7$ . Patients were not eligible if there were contraindications to thrombolysis or if the treating physician judged thrombolysis to be warranted as part of clinical standard of care. Randomization in the trial was completed by a computer-generated minimization algorithm to ensure balance for key variables, including age, sex, baseline NIHSS score, and time from symptom onset to randomization.

For this study, we compared efficacy and safety outcomes in patients assigned to tenecteplase vs standard of care in the prespecified subgroups of those  $>80$  years and  $\leq 80$  years, in the intention-to-treat population. The primary efficacy outcome was the same as that in the main trial, namely a responder analysis of the mRS at 90 days. This was defined as follows: (1) if the premorbid mRS was 0–1, then mRS 0–1 at 90 days was a responder or (2) if the premorbid mRS is 2, then mRS 0–2 was a responder. We also examined mRS 0–2 at 90 days, return to baseline function (i.e., return to premorbid mRS), and the full range of ordinal mRS outcomes. Short-term neurologic recovery was determined based on an NIHSS score of 0 at day 5 or on the day of hospital discharge (whichever is earlier). In patients with vessel occlusion on baseline CT angiography, we reported the proportion of early recanalization on the follow-up intracranial vascular imaging at 4–8 hours after randomization. A repeat CT angiogram was not required for patients enrolled based on symptomatic perfusion defects. Other secondary efficacy outcomes included the Lawton Instrumental Activities of Daily Living (IADL) Scale at 90 days (scored out of a total of 23 points using the method of the Comprehensive Assessment of Neurodegeneration and Dementia study)<sup>14</sup> and quality of life measured at 90 days on the EuroQol-5 Dimension score (EQ-5D-5L) where raw scores were converted into an index using country-specific tariffs. Safety outcomes collected out to 90 days included serious adverse events (SAEs), stroke progression, stroke recurrence, all-cause mortality, intracranial hemorrhage, extracranial hemorrhage, and other adverse events.

### Standard Protocol Approvals, Registrations, and Patient Consents

The trial was sponsored by the University of Calgary, Alberta, Canada. The trial was regulated by Health Canada and in the countries of each site as required. Local ethics board approval was obtained and all participants or their representative provided informed consent before enrollment. The trial was registered on ClinicalTrials.gov (NCT02398656).

### Statistical Analysis Plan

Data on baseline characteristics were summarized using frequencies and proportions for categorical variables and median with interquartile range for continuous variables. Outcomes were compared between  $\leq 80$  and  $>80$  years age groups in

unadjusted and adjusted analyses. Effect modification of age category ( $\leq 80$  and  $>80$  years) on the relationship between treatment arm and outcomes was assessed using 2-way multiplicative interaction terms (treatment  $\times$  age) in the multivariable models. Mixed-effects Poisson regressions were used for binary response variables such as the primary outcome of responder analysis, with no offset, adjusting for sex, prestroke mRS, baseline stroke severity (NIHSS), onset to randomization time, and treatment assignment (tenecteplase or standard of care) as fixed effects and region as random-effects to account for clustering of data within region, and with the robust Huber-Sandwich estimator of variance. Generalized ordered logistic regression was used to analyze ordinal mRS score. Health-related quality of life (EQ-5D index) was analyzed using mixed-effects linear regression models, adjusted as above. Adverse events including mortality were analyzed using the Cox regression model. All reported  $p$ -values were 2-sided, with  $p < 0.05$  considered statistically significant.

### Data Availability

Data collected for the study, including deidentified individual participant data and a data dictionary defining each field in the set, can be made available to others on reasonable request and after signing appropriate data sharing agreements and approval by all the respective ethics boards and appropriate data custodians.

## Results

Between April 27, 2015, and January 19, 2024, 886 patients were enrolled across 48 sites in Australia, Austria, Brazil, Canada, Finland, Ireland, New Zealand, Singapore, Spain, and the United Kingdom.<sup>12</sup> There were 432 patients (49%) assigned to tenecteplase and 454 (51%) to standard of care, including dual antiplatelet therapy ( $n = 262$ ), single antiplatelet therapy ( $n = 155$ ), and anticoagulation ( $n = 27$ ). Two patients withdrew consent and 2 patients were lost to follow-up. Among 884 patients in the intention-to-treat analysis, 208 (23.5%) were aged  $>80$  years, 106 (51.0%) of whom were assigned to tenecteplase. There were 365 (41.6%) female patients, of whom 115 were  $>80$  years. Comparing the baseline characteristics of patients assigned to tenecteplase vs standard of care among patients aged  $>80$  years and  $\leq 80$  years (Table 1), patients  $>80$  years assigned to tenecteplase were significantly older (median 86 years, interquartile range [IQR] 84–89 vs 84, IQR 83–88), less likely to have baseline/prestroke mRS 0 (59 [55.7%] vs 71 [69.6%]) and had shorter onset to treatment times than those assigned to standard of care (median 298 minutes, IQR 155–460 vs 330 minutes, IQR 204–492); no other significant differences were observed. Among patients aged  $\leq 80$  years, baseline characteristics were similar among those assigned to tenecteplase or standard of care.

A significant interaction between age and treatment was observed for mRS responder ( $p = 0.042$ ), mRS 0–1 ( $p = 0.011$ ), return to baseline function ( $p = 0.048$ ), EQ-5D index

**Table 1** Baseline Characteristics in Patients Enrolled Who Were ≤80 Years Compared With >80 Years

|                                   | Age ≤80 (n = 676) |               | Age >80 (n = 208) |               |
|-----------------------------------|-------------------|---------------|-------------------|---------------|
|                                   | Tenecteplase      | Control       | Tenecteplase      | Control       |
| N (%)                             | 326 (48.2)        | 350 (51.8)    | 106 (51.0)        | 102 (49.0)    |
| Demographics                      |                   |               |                   |               |
| Age (y)                           | 67 (59–75)        | 68 (58–74)    | 86 (84–89)*       | 84 (83–88)*   |
| Female sex                        | 126 (38.7)        | 127 (36.3)    | 62 (58.5)         | 53 (52.0)     |
| Weight (kg)                       | 80 (69–91)        | 80 (69–90)    | 69 (60–77)        | 68 (61–78)    |
| Weight (kg) <60                   | 26 (8.0)          | 29 (8.5)      | 26 (24.5)         | 23 (22.8)     |
| Race                              |                   |               |                   |               |
| Caucasian                         | 276 (84.7)        | 292 (83.4)    | 95 (89.6)         | 90 (88.2)     |
| First Nations                     | 4 (1.2)           | 5 (1.4)       |                   |               |
| Pacific Islander                  | 1 (0.3)           | 0 (0.0)       |                   |               |
| Asian                             | 32 (9.8)          | 34 (9.7)      | 8 (7.5)           | 8 (7.8)       |
| Black                             | 6 (1.8)           | 6 (1.7)       | 0 (0.0)           | 1 (1.0)       |
| Other                             | 7 (2.1)           | 13 (3.7)      | 3 (2.8)           | 3 (2.9)       |
| Ethnicity: Hispanic               | 4 (1.2)           | 8 (2.3)       | 1 (0.9)           | 1 (1.0)       |
| Medical history                   |                   |               |                   |               |
| Hypertension                      | 180 (55.2)        | 186 (53.1)    | 85 (80.2)         | 75 (73.5)     |
| Past smoking                      | 141 (43.3)        | 149 (42.6)    | 31 (29.2)         | 27 (26.5)     |
| Hyperlipidemia                    | 134 (41.1)        | 124 (35.4)    | 46 (43.4)         | 48 (47.1)     |
| Diabetes mellitus                 | 65 (19.9)         | 66 (18.9)     | 17 (16.0)         | 20 (19.6)     |
| Past stroke                       | 45 (13.8)         | 63 (18.0)     | 27 (25.5)         | 22 (21.6)     |
| Atrial fibrillation               | 57 (17.5)         | 54 (15.4)     | 34 (32.1)         | 24 (23.5)     |
| Ischemic heart disease            | 45 (13.8)         | 56 (16.0)     | 24 (22.6)         | 17 (16.7)     |
| Congestive heart failure          | 10 (3.1)          | 10 (2.9)      | 6 (5.7)           | 8 (7.8)       |
| Chronic renal failure             | 14 (4.3)          | 9 (2.6)       | 8 (7.5)           | 8 (7.8)       |
| Peripheral vascular disease       | 9 (2.8)           | 12 (3.4)      | 4 (3.8)           | 3 (2.9)       |
| Past ICH                          | 0 (0.0)           | 1 (0.3)       | 3 (2.8)           | 0 (0.0)       |
| Baseline mRS                      |                   |               |                   |               |
| mRS 0                             | 277 (85.0)        | 283 (80.9)    | 59 (55.7)*        | 71 (69.6)*    |
| mRS 1                             | 36 (11.0)         | 46 (13.1)     | 25 (23.6)         | 13 (12.7)     |
| mRS 2                             | 13 (4.0)          | 21 (6.0)      | 22 (20.8)         | 18 (17.6)     |
| Clinical presentation             |                   |               |                   |               |
| NIHSS score at baseline           | 2 (1–3)           | 2 (1–3)       | 2 (1–3)           | 2 (1–3)       |
| Hemoglobin (g/L)                  | 142 (131–152)     | 144 (134–154) | 132 (126–141)     | 133 (127–144) |
| Glucose (mM)                      | 6 (6–7)           | 6 (6–7)       | 6 (5–7)           | 6 (6–7)       |
| Creatinine (mM)                   | 82 (70–97)        | 83 (68–95)    | 84 (68–101)       | 87 (73–106)   |
| ASPECTS baseline                  | 10 (9–10)         | 10 (9–10)     | 10 (9–10)         | 10 (10–10)    |
| Onset-to-randomization time (min) | 284 (169–444)     | 272 (162–451) | 289 (140–431)     | 282 (173–432) |

Continued

**Table 1** Baseline Characteristics in Patients Enrolled Who Were ≤80 Years Compared With >80 Years (*continued*)

|                                                                                    | Age ≤80 (n = 676) |               | Age >80 (n = 208) |                |
|------------------------------------------------------------------------------------|-------------------|---------------|-------------------|----------------|
|                                                                                    | Tenecteplase      | Control       | Tenecteplase      | Control        |
| Onset-to-hospital arrival time (min)                                               | 147 (79–335)      | 146 (70–340)  | 148 (66–322)      | 174 (78–319)   |
| Onset-to-treatment time (min)                                                      | 292 (175–452)     | 302 (176–495) | 298 (155–460)*    | 330 (204–492)* |
| <b>Occlusion location at baseline: core laboratory</b>                             |                   |               |                   |                |
| Large vessel occlusion (intracranial ICA, M1-MCA)                                  | 36 (11.1)         | 42 (12.1)     | 17 (16.0)         | 8 (7.8)        |
| Medium vessel occlusion (M2-MCA or distal, A2-ACA or distal)                       | 175 (53.8)        | 191 (54.9)    | 60 (56.6)         | 54 (52.9)      |
| Vertebrobasilar circulation (VA, BA or branches, PCA)                              | 19 (5.8)          | 20 (5.7)      | 1 (0.9)           | 5 (4.9)        |
| Indirect occlusion (focal perfusion lesion on CTP but no visible occlusion on CTA) | 90 (27.7)         | 94 (27.0)     | 28 (26.4)         | 33 (32.4)      |
| No occlusion detected                                                              | 5 (1.5)           | 1 (0.3)       | 0 (0.0)           | 2 (2.0)        |

Abbreviations: ACA = anterior cerebral artery; ASPECTS = Alberta Stroke Program Early CT score; BA = basilar artery; CTA = computed tomography angiography; CTP = computed tomography perfusion; ICH = intracranial hemorrhage; ICA = internal carotid artery; IQR = interquartile range; MCA = middle cerebral artery; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; PCA = posterior cerebral artery; VA = vertebral artery. All numbers for continuous variables are reported as median (IQR). Significant differences between the tenecteplase and control arms are mentioned in the Results and indicated with an asterisk (\*).

( $p = 0.004$ ), and vessel recanalization ( $p < 0.001$ ). Hence, we assessed treatment effects across categories of age. Examining the primary outcome of mRS responder analysis (Figure 1, eTable 1), while there was no significant difference between tenecteplase and standard of care recipients in either age group, patients aged >80 years trended towards worse outcomes with tenecteplase (age >80 years: 57 [53.8%] with tenecteplase vs 70 [68.6%] with control, adjusted risk ratio [aRR] 0.82, 95% CI 0.65–1.04; age ≤80 years: 252 [77.3%] vs 268 [76.6%], aRR 1.00, 95% CI 0.97–1.03). Patients aged >80 years were significantly less likely to achieve mRS 0–1 when assigned to tenecteplase (49 [46.2%] vs 61 [59.8%], aRR 0.83, 95% CI 0.72–0.97), but there was no significant difference in patients ≤80 years (249 [76.4%] vs 260 [74.7%], aRR 1.01, 95% CI 0.98–1.03). There was no significant difference in achievement of mRS 0–2 for patients >80 years when assigned to tenecteplase (70 [66.0%] vs 82 [80.4%], aRR 0.83, 95% CI 0.67–1.02), whereas patients ≤80 years were less likely to achieve mRS 0–2 (282 [86.5%] vs 309 [88.8%], aRR 0.97, 95% CI 0.96–0.97). When examining the full range of 90-day mRS scores in both age groups (Figure 2), there was no significant difference in adjusted common odds ratio (acOR) with tenecteplase vs standard care in patients >80 years (acOR 1.50, 95% CI 0.89–2.52) and in those ≤80 years, although the direction of effect was different (acOR 0.91, 95% CI 0.69–1.21). Return to baseline function was achieved by 43 (40.6%) patients >80 years in the tenecteplase arm and 49 (48.0%) in the standard care arm (aRR 0.82, 95% CI 0.58–1.16); rates were numerically higher in the tenecteplase arm in patients ≤80 years, but the difference was not significant (169 [51.8%] vs 173 [49.4%], aRR 1.07, 95% CI 0.92–1.24).

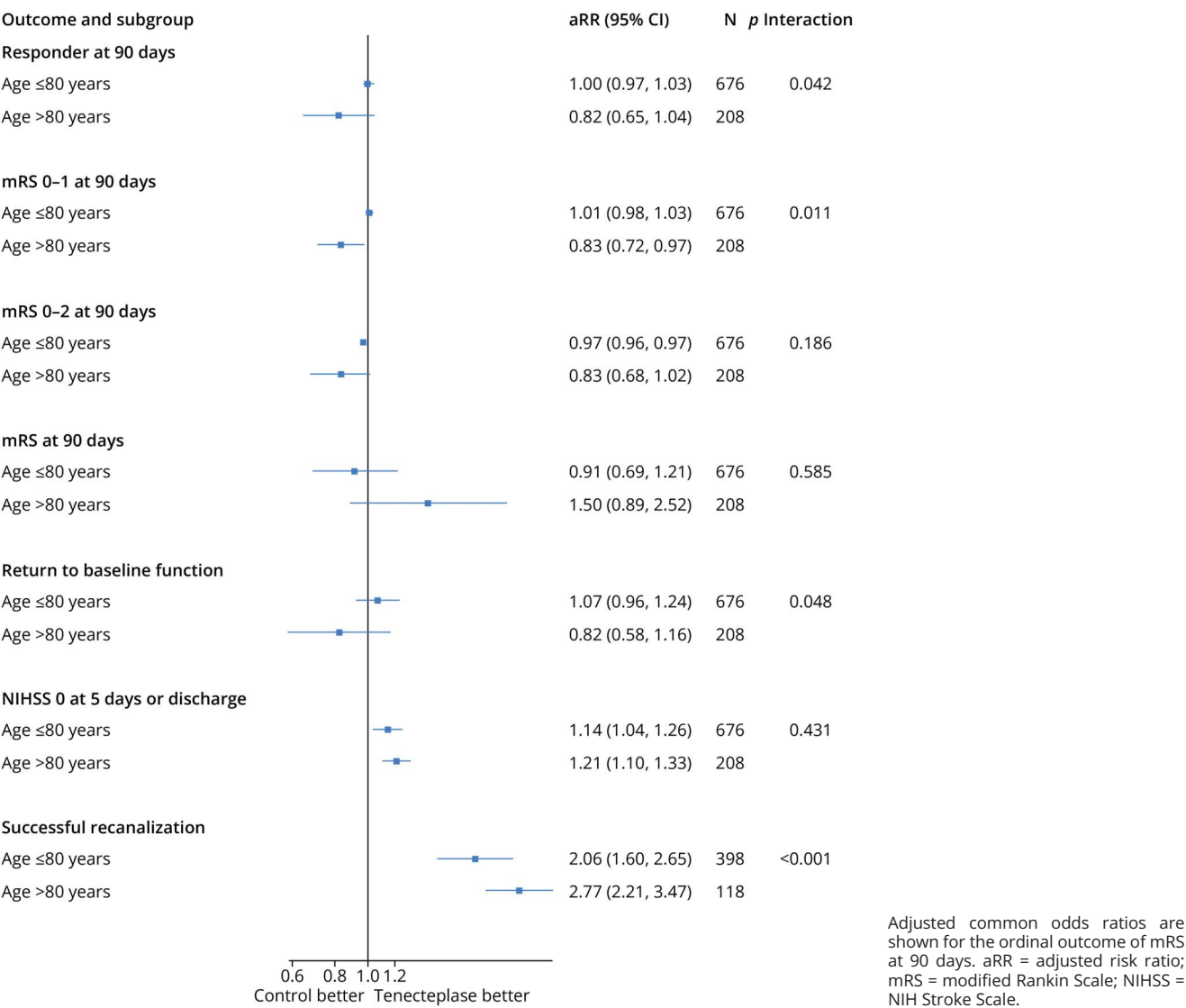
In both age groups, patients with visible occlusions were more than twice as likely to achieve recanalization on follow-up imaging when assigned to tenecteplase (>80 years: 30 [46.9%] vs

9 [16.7%], aRR 2.77, 95% CI 2.21–3.47; ≤80 years: 92 [47.4%] vs 47 [23.0%], aRR 2.06, 95% CI 1.60–2.65). Patients assigned to tenecteplase were also more likely to achieve NIHSS 0 at 5 days/discharge (e.g., >80 years: 56 [53.3%] vs 45 [44.1%], aRR 1.21, 95% CI 1.10–1.33; ≤80 years: 191 [59.3%] vs 181 [51.7%], aRR 1.14, 95% CI 1.04–1.26).

There was no significant difference in reported performance on instrumental activities of daily living in the Lawton IADL scale between tenecteplase and standard care arms in both age groups (>80 years: adjusted difference –1.05, 95% CI –2.63 to 0.53; ≤80 years: adjusted difference –0.02, 95% CI –0.49 to 0.45, Figure 3). Examining patient-reported quality of life, the EQ-5D index scores were significantly lower among patients >80 years who were assigned to tenecteplase vs standard care (adjusted difference –0.08, 95% CI –0.12 to –0.04), but there was no significant difference in patients ≤80 years (adjusted difference –0.02, 95% CI –0.05 to 0.00). There was no significant difference in EQ-5D visual analog scale ratings between tenecteplase and standard care arms in patients >80 years (adjusted-difference –3.67, 95% CI –11.33 to 4.00), but ratings were significantly lower among patients ≤80 years assigned to tenecteplase (adjusted-difference –3.53, 95% CI –5.85 to –1.22).

As for safety outcomes (Table 2), 90-day mortality was higher among patients ≤80 years in the tenecteplase arm compared with standard of care (14 [4.3%] vs 3 [0.9%], hazard ratio [HR] 4.37, 95% CI 1.23–15.49), but this was not significant among patients >80 years (6 [5.7%] vs 2 [2.0%], HR 2.94, 95% CI 0.59–14.58). SAEs (Table 3) were more frequent in the tenecteplase arm compared with standard of care among patients >80 years, but not significantly so among those ≤80 years (≤80 years: risk ratio [RR] 1.11, 95% CI 0.82–1.49, >80 years: 2.29, 95% CI 1.27–4.13). Symptomatic

**Figure 1** Efficacy Outcomes in Patients Treated With Tenecteplase vs Standard of Care in Patients  $\leq 80$  Years and  $>80$  Years Showing aRRs



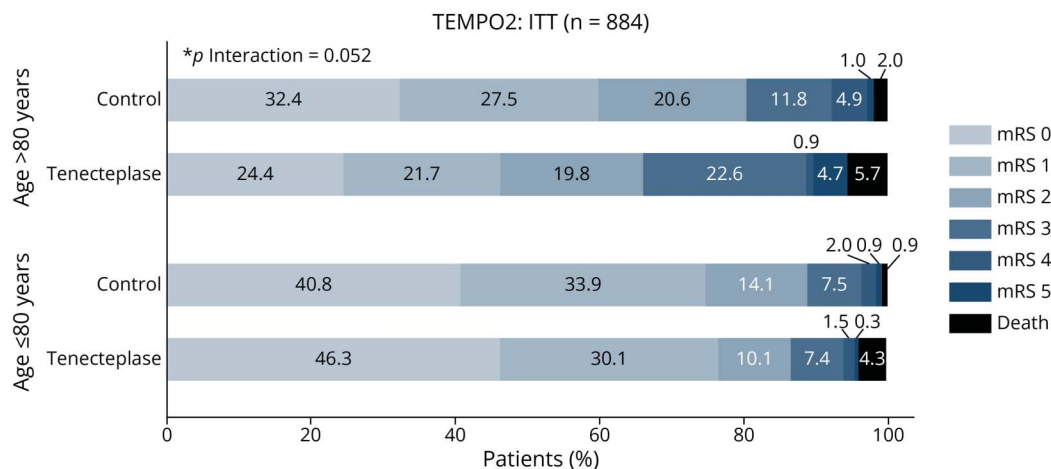
intracranial hemorrhage occurred in only 1 patient  $>80$  years, who received tenecteplase. Extracranial bleeding was, however, significantly more common among patients  $>80$  years who received tenecteplase vs standard of care (4 [3.8%] vs 0,  $p = 0.048$ ), and thus, these patients had significantly more hemorrhagic SAEs overall (5 [4.7%] vs 0,  $p = 0.026$ ). Stroke progression and recurrent stroke, when considered in combination, were also significantly more frequent in the tenecteplase arm among patients aged  $>80$  years (16 [15.1%] vs 6 [5.9%], RR 2.57, 95% CI 1.05–6.30).

### Classification of Evidence

This study provides Class III evidence that in patients  $>80$  years with minor stroke and visible occlusion or perfusion lesion, IV thrombolysis with tenecteplase is associated with worse functional outcomes at 90 days compared with non-thrombolytic standard of care therapies.

### Discussion

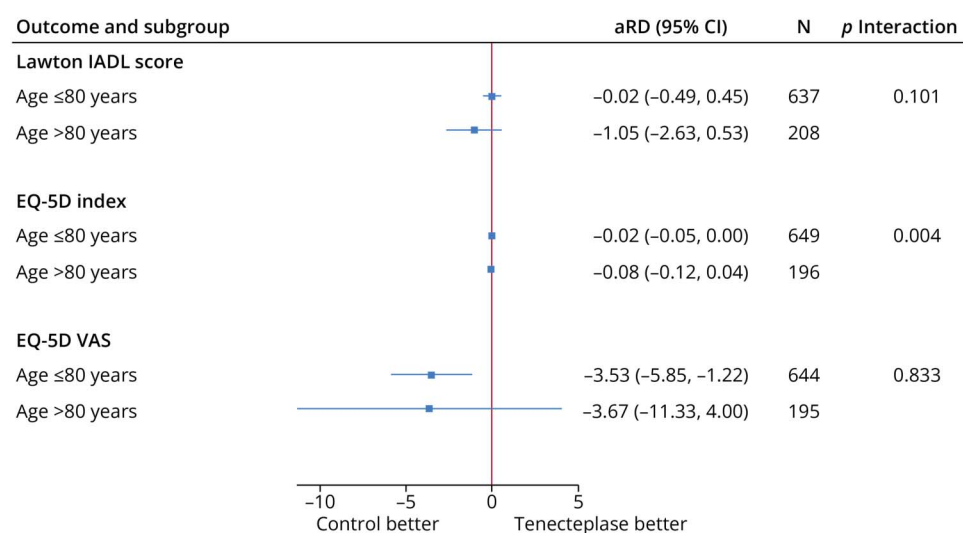
In this post hoc analysis of the TEMPO-2 randomized-controlled trial, we found that patients  $>80$  years with minor ischemic stroke and a symptomatic intracranial occlusion or focal perfusion lesion achieved worse functional and quality of life outcomes when assigned to tenecteplase vs standard of care, despite achieving higher rates of recanalization and early neurologic recovery according to the NIHSS, while this was not the case for those aged  $\leq 80$ . Patients  $>80$  years assigned to tenecteplase also had higher rates of SAEs compared with those assigned to standard of care, particularly extracranial hemorrhage and stroke progression/recurrence. These findings help illuminate the complex trade-offs of risks and benefits that are involved when considering IV thrombolysis in older patients with minor stroke.

**Figure 2** Distribution of 90-Day mRS in Patients  $\leq 80$  Years and  $>80$  Years of Age in the TEMPO-2 Trial, Each Stratified by Tenecteplase vs Control Assignment According to ITT

ITT = intention to treat; mRS = modified Rankin Scale.

The benefit of IV thrombolysis with alteplase for major acute ischemic stroke has been well demonstrated in older patients. A pooled analysis of individual patient data from 7 randomized-controlled trials of alteplase (0.9 mg/kg) vs placebo or open control found that among patients aged  $>80$  years, alteplase (n = 518) vs placebo (n = 510) was associated with a higher proportion of mRS 0–1 at 90 days (19.1% vs 13.1%) and similar 90-day mortality (29.5% vs 30.2%), with the odds of a favorable mRS outcome being independent of age.<sup>15</sup> This was further supported by data from the European Safe Implementation of Thrombolysis in Upper Time Window Monitoring Study registry, which found that mRS 0–1 was achieved by 26.6% of 1,929

patients  $>80$  years who received alteplase in routine practice<sup>15</sup>; a similar rate of mRS 0–1 of 26% was observed among 270 patients  $\geq 80$  years who received alteplase in the Canadian Alteplase for Stroke Effectiveness Study.<sup>16</sup> In our study, among 208 patients  $>80$  years with minor ischemic stroke, we unsurprisingly found better outcomes than in these prior analyses, with the majority of patients achieving favorable functional outcomes on the mRS regardless of how they were defined (responder analysis, mRS 0–1, mRS 0–2, or return to baseline), and with far lower mortality (5.7% with tenecteplase and 2.0% with standard care, in contrast to 29.9% in the aforementioned analysis). These results are broadly comparable to the rates of mRS 0–1 at 3-months (53%–57%)

**Figure 3** Efficacy Outcomes in Patients Treated With Tenecteplase vs Standard of Care in Patients  $\leq 80$  Years and  $>80$  Years Showing aRD

aRD = adjusted risk difference; EQ-5D = EuroQol-5 Dimension; IADL = Instrumental Activities of Daily Living; VAS = Visual Analog Scale.

**Table 2** Safety Outcomes in Patients Treated With Tenecteplase vs Standard of Care in  $\leq 80$  Years and  $>80$  Years

| End points                                          | Age $\leq 80$ (n = 676) |            |                          | Age $>80$ (n = 208) |            |                          |
|-----------------------------------------------------|-------------------------|------------|--------------------------|---------------------|------------|--------------------------|
|                                                     | Tenecteplase            | Control    | Unadjusted difference    | Tenecteplase        | Control    | Unadjusted difference    |
| <b>N (%)</b>                                        | 326 (48.2)              | 350 (51.8) |                          | 106 (51.0)          | 102 (49.0) |                          |
| <b>Death within 90 d, n (%)</b>                     | 14 (4.3)                | 3 (0.9)    | HR: 4.37 (1.23 to 15.49) | 6 (5.7)             | 2 (2.0)    | HR: 2.94 (0.59 to 14.58) |
| <b>Serious adverse event, n (%)</b>                 | 69 (21.2)               | 67 (19.1)  | RR: 1.11 (0.82 to 1.49)  | 31 (29.2)           | 13 (12.7)  | RR: 2.29 (1.27 to 4.13)  |
| <b>Stroke progression, n (%)</b>                    | 23 (7.1)                | 28 (8.0)   | RR: 0.88 (0.52 to 1.50)  | 12 (11.3)           | 5 (4.9)    | RR: 2.31 (0.84 to 6.32)  |
| <b>Stroke recurrence, n (%)</b>                     | 12 (3.7)                | 14 (4.0)   | RR: 0.92 (0.43 to 1.96)  | 4 (3.8)             | 1 (1.0)    | RR: 3.85 (0.44 to 33.86) |
| <b>Symptomatic intracranial hemorrhage, n (%)</b>   | 7 (2.1)                 | 2 (0.6)    | RR: 3.76 (0.79 to 17.96) | 1 (0.9)             | 0 (0.0)    | NA                       |
| <b>ECASS radiologic hemorrhage type</b>             |                         |            |                          |                     |            |                          |
| <b>None</b>                                         | 276 (84.7)              | 318 (90.9) |                          | 94 (92.2)           | 94 (88.7)  |                          |
| <b>HI1</b>                                          | 20 (6.1)                | 14 (4.0)   |                          | 3 (2.8)             | 5 (4.9)    |                          |
| <b>HI2</b>                                          | 9 (2.8)                 | 11 (3.1)   |                          | 5 (4.7)             | 0 (0.0)    |                          |
| <b>PH1</b>                                          | 4 (1.2)                 | 1 (0.3)    |                          | 0                   | 0          |                          |
| <b>PH2</b>                                          | 9 (2.8)                 | 2 (0.6)    |                          | 2 (1.9)             | 0 (0.0)    |                          |
| <b>rPH</b>                                          | 6 (1.8)                 | 0 (0.0)    |                          | 2 (1.9)             | 3 (2.9)    |                          |
| <b>SAH</b>                                          | 2 (0.6)                 | 4 (1.1)    |                          | 0                   | 0          |                          |
| <b>Any hemorrhage on follow-up imaging, n (%)</b>   | 50 (15.4)               | 32 (9.5)   | RR: 1.63 (1.07 to 2.47)  | 12 (11.3)           | 8 (8.2)    | RR: 1.39 (0.59 to 3.25)  |
| <b>Rescue thrombectomy for index stroke, n (%)</b>  | 12 (3.7)                | 10 (2.9)   | RR: 1.29 (0.56 to 2.94)  | 3 (2.8)             | 0 (0.0)    | NA                       |
| <b>Length of stay in hospital (d), median (IQR)</b> | 3 (2–7)                 | 3 (2–6)    | RD: 0 (–0.68 to 0.68)    | 4 (2–11)            | 4 (2–7)    | RD: 1 (–0.67 to 2.67)    |

Abbreviations: ECASS = European Cooperative Acute Stroke Study; HI = hemorrhagic infarction; PH = parenchymal hematoma; rPH = remote parenchymal hematoma; NA = not applicable; RD = risk difference; RR = risk ratio; SAH = subarachnoid hemorrhage.

observed in the subset of patients with minor stroke enrolled in the Norwegian Tenecteplase Stroke Trial comparing tenecteplase with alteplase.<sup>17</sup> However, return to prestroke mRS (i.e., baseline function) occurred only in a minority of patients  $>80$  years in TEMPO-2 (44% overall), underscoring the functional impact that these minor strokes can have on patients beyond what is measured by conventional definitions of favorable mRS outcomes.<sup>18</sup> This was also reflected by the median EQ-5D Visual Analog Scale ratings of 75 of 100 for these patients.

Importantly, in contrast to studies of thrombolysis in older individuals for major stroke, we found that functional outcomes did not improve with tenecteplase vs standard of care among patients  $>80$  years and were significantly worse for the secondary outcome of mRS 0–1. Whereas older patients, particularly those in the tenecteplase arm, generally had lower rates of premorbid mRS 0, this observation of worse mRS 0–1 outcome with treatment remained significant after adjusting for prestroke mRS in addition to other factors. In addition, whereas the differences between tenecteplase and standard care arms were not significant among patients  $>80$  years for the primary responder analysis, mRS 0–2, return to baseline, or ordinal analysis of the mRS, responder rates and rates of

mRS 0–2 and return to baseline were numerically lower in those in the tenecteplase arm, as was the median mRS. Thus, similar to the PRISMS, ARAMIS, PUMICE, TRUST, and overall TEMPO-2 results, patients  $>80$  years in TEMPO-2 also did not benefit from thrombolysis compared with standard of care with respect to functional outcomes. The significant interactions observed in our analysis between age group and treatment for our primary mRS responder outcome and several secondary outcomes stand in contrast to the results of the PRISMS and ARAMIS trials of IV alteplase compared with aspirin and dual antiplatelet therapy, respectively, and the recent PUMICE trial of IV prourokinase compared with nonthrombolytic standard care, which all found no significant interaction between age and treatment assignment.<sup>8–10</sup> However, the PRISMS and ARAMIS trials dichotomized age as  $\geq 65$  years vs  $< 65$  years, so did not specifically examine outcomes in very older patients, and all 3 trials only examined age-associated differences in the primary outcome of mRS 0–1 at 90 days.

In contrast to the poorer functional outcomes among patients  $>80$  years with tenecteplase, we found, as in the overall trial, that patients receiving tenecteplase had higher rates of recanalization of occlusions as well as higher rates of

**Table 3** Breakdown of Serious Adverse Events in Patients Aged  $\leq 80$  Years vs  $>80$  Years

|                                                         | Age $\leq 80$ (n = 676) |                  | Age $>80$ (n = 208)   |                  |
|---------------------------------------------------------|-------------------------|------------------|-----------------------|------------------|
|                                                         | Tenecteplase<br>n (%)   | Control<br>n (%) | Tenecteplase<br>n (%) | Control<br>n (%) |
| <b>N (%)</b>                                            | 326 (48.2)              | 350 (51.8)       | 106 (51.0)            | 102 (49.0)       |
| <b>Any serious adverse event</b>                        | 69 (21.2)               | 67 (19.1)        | 31 (29.2)             | 13 (12.7)        |
| <b>Symptomatic intracranial hemorrhage<sup>a</sup></b>  | 7 (2.1)                 | 2 (0.6)          | 1 (0.9)               | 0 (0.0)          |
| <b>Serious extracranial bleeding</b>                    | 2 (0.6)                 | 3 (0.9)          | 4 (3.8)               | 0 (0.0)          |
| <b>Any serious hemorrhage (sICH plus extra-cranial)</b> | 9 (2.8)                 | 5 (1.4)          | 5 (4.7)               | 0 (0.0)          |
| <b>Stroke progression</b>                               | 23 (7.1)                | 28 (8.0)         | 12 (11.3)             | 5 (4.9)          |
| <b>Recurrent stroke</b>                                 | 12 (3.7)                | 14 (4.0)         | 4 (3.8)               | 1 (1.0)          |
| <b>Stroke progression or recurrent stroke</b>           | 35 (10.7)               | 42 (11.9)        | 16 (15.1)             | 6 (5.9)          |
| <b>Transient ischemic attack</b>                        | 5 (1.5)                 | 4 (1.1)          | 0 (0.0)               | 0 (0.0)          |
| <b>Headache</b>                                         | 8 (2.5)                 | 3 (0.9)          | 0 (0.0)               | 0 (0.0)          |
| <b>Aspiration pneumonia</b>                             | 3 (0.9)                 | 2 (0.6)          | 3 (2.8)               | 0 (0.0)          |
| <b>Atrial fibrillation</b>                              | 2 (0.6)                 | 3 (0.9)          | 2 (1.9)               | 0 (0.0)          |
| <b>Congestive heart failure</b>                         | 2 (0.6)                 | 1 (0.3)          | 3 (2.8)               | 0 (0.0)          |
| <b>Seizure</b>                                          | 3 (0.9)                 | 3 (0.9)          | 0 (0.0)               | 0 (0.0)          |
| <b>Urinary tract infection</b>                          | 2 (0.6)                 | 3 (0.9)          | 0 (0.0)               | 1 (1.0)          |
| <b>Other</b>                                            | 10 (3.1)                | 8 (2.3)          | 6 (5.7)               | 6 (5.9)          |

<sup>a</sup>All symptomatic ICH occurred acutely in the first day. Only 1 extracranial bleeding event occurred on the first day. All other major bleeding events occurred late, days or weeks after treatment. On this basis, we do not believe that these bleeding events were related to tenecteplase. ICH = intracranial hemorrhage; sICH = symptomatic intracranial hemorrhage.

short-term neurologic recovery (NIHSS 0 at 5 days/discharge). The latter finding was in contrast to the results of a recent single-center observational study from Norway of patients  $>80$  years who underwent thrombolysis with alteplase for stroke within 4.5 hours and found that patients with minor stroke (NIHSS  $<3$ ) had worse short-term neurologic outcomes (based on mRS at day 7 or discharge). This was largely attributed to greater neurologic worsening after thrombolysis in that population. Interestingly, we did find that stroke progression was numerically more frequent in the tenecteplase arm among patients aged  $>80$  years although few patients were affected (n = 17 total); this is comparable with the observation of greater rates of early neurologic deterioration in the alteplase group vs the dual antiplatelet therapy group in the ARAMIS trial (9.1% vs 4.6%), in somewhat paradoxical contrast to the greater rates of early neurologic recovery found in the alteplase group in the per-protocol and as-treated analyses in that trial.<sup>8</sup> It is worth noting that the absolute difference in early neurologic recovery (NIHSS = 0 at 5 days/discharge) between treatment and control arms in TEMPO-2 was 9.2% in patients  $>80$  years, numerically higher than that of 7.6% in those  $\leq 80$  years. This was in striking contrast to the disparate predicted absolute benefits in neurologic improvement rates of  $>50\%$  in patients  $<70$  years vs  $<10\%$  in those  $>80$  years in a recent

multicenter retrospective study; these differences might relate to the limits of nonrandomized data in the previous study but also to the fact that older patients enrolled in TEMPO-2 still had to be high functioning.<sup>19</sup> Nevertheless, the improvements in recanalization and early neurologic recovery in our study did not translate into benefits in functional or quality of life outcomes, especially so in the older patients. The fact that older patients fared worse overall than patients  $\leq 80$  years is unsurprising, given the importance of age as a prognostic factor in stroke, and is also in agreement with the recent findings of the Alteplase compared with Tenecteplase trial,<sup>20</sup> in which patients  $\geq 80$  years (n = 543, generally with major stroke) treated with IV thrombolysis had significantly lower rates of 90-day mRS 0–1 and return to baseline status than those  $<80$  years (n = 1,034), as well as higher mortality and poorer quality of life outcomes.<sup>21</sup>

In considering the discrepancy between short-term and 90-day outcomes, which were especially striking in patients  $>80$  years in our study and ran counter to the strong agreement between early NIHSS and 90-day mRS outcomes observed in prior studies,<sup>22,23</sup> we examined safety outcomes in the study with particular interest. Indeed, we found that SAEs were over twice as frequent in the tenecteplase arm compared with standard care among patients  $>80$  years. Prior analyses of

thrombectomy trials have shown that apparent treatment success on short-term measures (such as post-treatment infarct volume) can be eroded by posttreatment complications related to index stroke evolution, secondary prevention, and quality of stroke unit care.<sup>24,25</sup> In our study, the differences in SAEs among older patients assigned to tenecteplase were partly accounted for by differences in hemorrhagic outcomes, most notably extracranial hemorrhages, which did not occur in the control arm among those >80 years, and also by stroke recurrence and progression. As most of the hemorrhagic events occurred late, days or weeks after treatment, it seems unlikely that they were caused by tenecteplase. The greater frequency of stroke recurrence and progression-related SAEs among older tenecteplase recipients in TEMPO-2—when also considered in the context of the more frequent early neurologic deterioration observed in alteplase recipients in the ARAMIS trial<sup>8</sup>—raises the important possibility that thrombolysis in minor stroke may carry an opportunity cost in the form of delayed antithrombotic initiation. Since antithrombotic treatments are typically withheld within 24 hours of thrombolysis, their delayed initiation may increase the risk of stroke recurrence or progression, given the known time-dependent efficacy of antiplatelet therapy<sup>26</sup>; older patients are also more likely to have early stroke recurrence as well as progression or early neurologic deterioration.<sup>27,28</sup>

The study had several limitations. First, the findings are based on subgroup analyses within the TEMPO-2 randomized-controlled trial which was not specifically powered to study thrombolysis outcomes in older patients, and therefore, our results should be interpreted cautiously. Second, the study did not explore some other potential factors such as stroke etiology or brain frailty that could influence outcomes.<sup>29</sup> Third, the patients >80 years in TEMPO-2 were by definition high-functioning older adults, having met the trial inclusion criteria including having a maximal premorbid mRS of 2; this limits the generalizability of our findings to older patients with stroke routinely encountered in clinical practice.<sup>30</sup> Fourth, although delayed initiation of antiplatelet therapy in the tenecteplase arm could have been a factor for the observed lack of benefit with thrombolysis at 90 days, we did not directly measure the time to initiation of antiplatelet therapy in this arm. Fifth, we did not have information on rehabilitation services received by patients; differences in this regard could also potentially contribute to observed discrepancies between favorable early NIHSS vs later mRS outcomes in this study.

In conclusion, older patients with minor stroke and visible occlusion or perfusion lesion assigned to tenecteplase in the TEMPO-2 randomized-controlled trial were more likely to achieve recanalization of visible occlusions and short-term neurologic recovery on the NIHSS, as were younger patients. However, patients aged >80 years assigned to tenecteplase did not benefit from thrombolysis for 90-day functional outcomes and experienced worse safety outcomes with more frequent hemorrhagic SAEs. Overall, our results, while post

hoc, do not support thrombolysis for minor stroke in the older population as compared with antiplatelet therapy alone.

## Author Contributions

A. Ganesh: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S. Vatanpour: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A.Y.X. Yu: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. P. Barber: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. P.M.C. Choi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. T.S. Field: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. M.P. Kate: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. T.J. Kleinig: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. M.C.F. Shamy: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. K. Chatterjee: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. C.A. Molina: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. B.C.V. Campbell: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. R. Appireddy: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. K.W. Muir: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M.D. Hill: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S.B. Coutts: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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## References

- Tinker RJ, Smith CJ, Heal C, et al. Predictors of mortality and disability in stroke-associated pneumonia. *Acta Neurol Belg*. 2021;121(2):379-385. doi:10.1007/s13760-019-01148-w
- Yu AYY, Fang J, Porter J, Austin PC, Smith EE, Kapral MK. Hospital-based cohort study to determine the association between home-time and disability after stroke by age, sex, stroke type and study year in Canada. *BMJ Open*. 2019;9(11):e031379. doi:10.1136/bmjopen-2019-031379
- Hirayama A, Goto T, Faridi MK, Camargo CA Jr, Hasegawa K. Age-related differences in the rate and diagnosis of 30-day readmission after hospitalization for acute ischemic stroke. *Int J Stroke*. 2018;13(7):717-724. doi:10.1177/1747493018772790
- Asdagh N, Butcher KS, Hill MD. Risks and benefits of thrombolysis in the elderly. *Int J Stroke*. 2012;7(2):142-149. doi:10.1111/j.1747-4949.2011.00744.x
- Lee HJ, Lee JS, Choi JC, et al. Simple estimates of symptomatic intracranial hemorrhage risk and outcome after intravenous thrombolysis using age and stroke severity. *J Stroke*. 2017;19(2):229-231. doi:10.5853/jos.2016.01109
- Powers WJ, Rabinstein AA, Ackerson T, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2019;50(12):e344-e418. doi:10.1161/STR.0000000000000211
- Singh N, Menon BK, Dmytriw AA, Regenshard RW, Hirsch JA, Ganesh A. Replacing alteplase with tenecteplase: is the time ripe? *J Stroke*. 2023;25(1):72-80. doi:10.5853/jos.2022.02880
- Chen HS, Cui Y, Zhou ZH, et al. Dual antiplatelet therapy vs alteplase for patients with minor nondisabling acute ischemic stroke: the ARAMIS randomized clinical trial. *JAMA*. 2023;329(24):2135-2144. doi:10.1001/jama.2023.7827
- Khatir P, Kleindorfer DO, Devlin T, et al. Effect of alteplase vs aspirin on functional outcome for patients with acute ischemic stroke and minor nondisabling neurologic deficits: the PRISMS randomized clinical trial. *JAMA*. 2018;320(2):156-166. doi:10.1001/jama.2018.8496
- Xiong Y, Meng X, Jin A, et al. Prourokinase vs standard care for patients with mild ischemic stroke: the PUMICE randomized clinical trial. *JAMA Neurol*. 2025;82(3):258-266. doi:10.1001/jamaneurol.2024.4688
- Tao Y, Gao Y, Zhao L, et al. Effect of intravenous urokinase vs best medicine treatment on functional outcome for patients with acute minor stroke (TRUST): a randomized controlled trial. *BMC Med*. 2025;23(1):6. doi:10.1186/s12916-024-03820-2
- Coutts SB, Ankolekar S, Appireddy R, et al. Tenecteplase versus standard of care for minor ischaemic stroke with proven occlusion (TEMPO-2): a randomised, open label, phase 3 superiority trial. *Lancet*. 2024;403(10444):2597-2605. doi:10.1016/S0140-6736(24)00921-8
- Singh N, Kenney CC, Butcher KS, et al. A randomized controlled trial of tenecteplase versus standard of care for minor ischemic stroke with proven occlusion (TEMPO-2): rationale and design of a multicenter, randomized open-label clinical trial. *Int J Stroke*. 2024;19(7):817-822. doi:10.1177/17474930241253702
- Chertkow H, Borrie M, Whitehead V, et al. The comprehensive assessment of neurodegeneration and dementia: Canadian cohort study. *Can J Neurol Sci*. 2019;46(5):499-511. doi:10.1017/cjn.2019.27
- Bluhmki E, Danays T, Biegert G, Hacke W, Lees KR. Alteplase for acute ischemic stroke in patients aged >80 years: pooled analyses of individual patient data. *Stroke*. 2020;51(8):2322-2331. doi:10.1161/STROKEAHA.119.028396
- Sylaja PN, Cote R, Buchan AM, Hill MD; Canadian Alteplase for Stroke Effectiveness Study CASES Investigators. Thrombolysis in patients older than 80 years with acute ischaemic stroke: Canadian Alteplase for Stroke Effectiveness Study. *J Neurol Neurosurg Psychiatry*. 2006;77(7):826-829. doi:10.1136/jnnp.2005.086595
- Ronning OM, Logallo N, Thommessen B, et al. Tenecteplase versus alteplase between 3 and 4.5 hours in low National Institutes of Health Stroke Scale. *Stroke*. 2019;50(2):498-500. doi:10.1161/STROKEAHA.118.024223
- Kapoor A, Lancot KL, Bayley M, et al. "Good outcome" isn't good enough: cognitive impairment, depressive symptoms, and social restrictions in physically recovered stroke patients. *Stroke*. 2017;48(6):1688-1690. doi:10.1161/STROKEAHA.117.016728
- Thomas T, Dubinski MJ, Elgendy O, et al. Age-dependent effect of thrombolytics on likelihood of National Institutes of Health Stroke Scale improvement in minor stroke. *Ibrain*. 2025;11(4):504-511. doi:10.1002/ibra.70008
- Menon BK, Buck BH, Singh N, et al. Intravenous tenecteplase compared with alteplase for acute ischaemic stroke in Canada (AcT): a pragmatic, multicentre, open-label, registry-linked, randomised, controlled, non-inferiority trial. *Lancet*. 2022;400(10347):161-169. doi:10.1016/S0140-6736(22)01054-6
- Singh N, Bala F, Ademola A, et al. Safety, functional disability, healthcare utilization, and quality-of-life outcomes in elderly receiving alteplase and tenecteplase: a secondary analysis from the AcT trial. *Stroke*. 2025;56(5):1169-1179. doi:10.1161/STROKEAHA.124.049512
- Rinkel LA, Ospel JM, Kappellhof M, et al. Comparing early National Institutes of Health Stroke Scale versus 90-day modified Rankin scale outcomes in acute ischemic stroke trials: a systematic review and analysis. *J Am Heart Assoc*. 2025;14(9):e040304. doi:10.1161/JAHA.124.040304
- Saver JL, Altman H. Relationship between neurologic deficit severity and final functional outcome shifts and strengthens during first hours after onset. *Stroke*. 2012;43(6):1537-1541. doi:10.1161/STROKEAHA.111.636928
- Ganesh A, Menon BK, Assis ZA, et al. Discrepancy between post-treatment infarct volume and 90-day outcome in the ESCAPE randomized controlled trial. *Int J Stroke*. 2021;16(5):593-601. doi:10.1177/1747493020929943
- Ganesh A, Ospel JM, Menon BK, et al. Assessment of discrepancies between follow-up infarct volume and 90-day outcomes among patients with ischemic stroke who received endovascular therapy. *JAMA Netw Open*. 2021;4(11):e2132376. doi:10.1001/jamanetworkopen.2021.32376
- Rothwell PM, Algra A, Chen Z, Diener H-C, Norrving B, Mehta Z. Effects of aspirin on risk and severity of early recurrent stroke after transient ischaemic attack and ischaemic stroke: time-course analysis of randomised trials. *Lancet*. 2016;388(10042):365-375. doi:10.1016/S0140-6736(16)30468-8
- Kolmos M, Christoffersen L, Kruuse C. Recurrent ischemic stroke: a systematic review and meta-analysis. *J Stroke Cerebrovasc Dis*. 2021;30(8):105935. doi:10.1016/j.jstrokecerebrovasdis.2021.105935
- Pensato U, Coutts SB, van Adel B, et al. Incidence, associations, and mechanisms of unexplained early neurologic deterioration after thrombectomy in stroke patients. *Neurology*. 2025;105(7):e214261. doi:10.1212/WNL.000000000000214261
- Benali F, Singh N, Fladt J, et al. Mediation of age and thrombectomy outcome by neuroimaging markers of frailty in patients with stroke. *JAMA Netw Open*. 2024;7(1):e2349628. doi:10.1001/jamanetworkopen.2023.49628
- Ganesh A, Luengo-Fernandez R, Pendlebury ST, Rothwell PM. Long-term consequences of worsened poststroke status in patients with premorbid disability. *Stroke*. 2018;49(10):2430-2436. doi:10.1161/STROKEAHA.118.022416